LONGMAN GCSE BIOLOGY WORKSHEETS

5: Natural selection

This worksheet uses some invented data to test your understanding of the principles of natural selection. It also deals with some aspects of ecology.

A farmer's grain crop is under attack from an insect pest. This pest, the corn weevil, bores holes in the seeds of the crop in summer and lays its eggs. The adults then die. The eggs hatch into larvae, which feed from the seeds until they mature and are ready to fly off. The adult weevils feed from a different food plant. They mate and then hibernate through the winter, starting the cycle again the next year. For every adult weevil, an average of 5 offspring are produced every year.

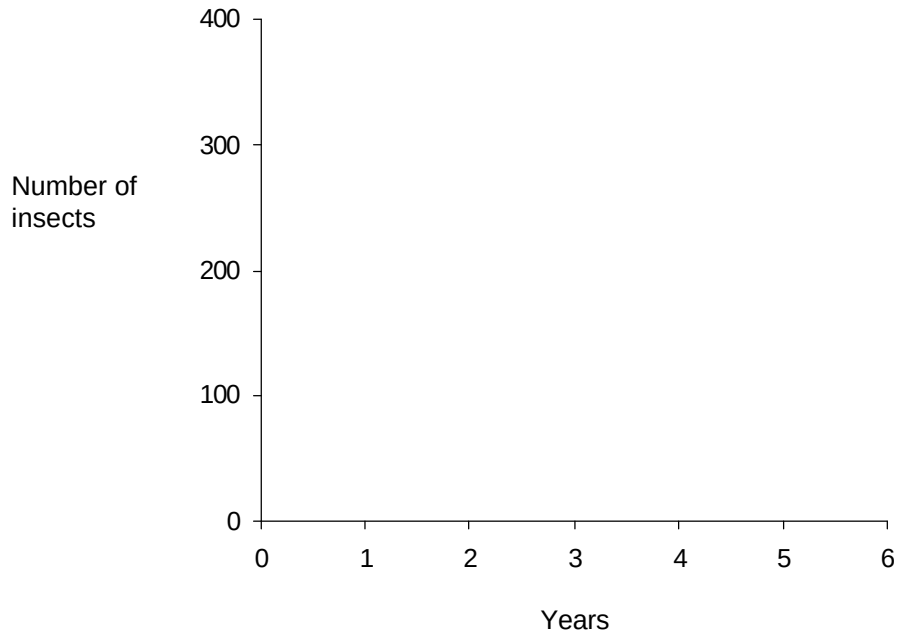
The farmer has tried to control the corn weevil by spraying his crops in summer with an insecticide called 'Bugloss'. When he first used this, it was very successful. Only 15% of the immature weevils survived the application of Bugloss. However, in the last year of spraying, 25% survived. The corn weevil is known to develop a mutation which makes it resistant to Bugloss, and the farmer has been advised that resistance has developed in the pest attacking his crop. This resistance is inherited.

Assuming that weevils that are resistant to Bugloss show a 25% survival rate following spraying, while non-resistant weevils show a 15% survival rate, and that both resistant and non-resistant weevils produce 5 offspring per year, answer the following questions:

Questions

1. Starting with 100 insects, calculate the number of weevils you would expect to find each year for six successive years, for (a) resistant insects and (b) non-resistant insects (assuming the survival rate of the insects stays constant at 25% and 15% each year).

2. a) Plot these numbers on a graph, using the same axes for both types of insect:



b) Comment on the shape of the graph.

3. Explain the terms **mutation** and **resistance** (to insecticide).

4. Darwin' theory says that evolution takes place as a result of the process of **natural selection**, through **survival of the fittest**. With reference to your calculations and the graph, explain how the example of the corn weevil illustrates these processes.

5. If the farmer continues to spray with bugloss, he will be wasting his money. Describe another way he could reduce the numbers of the corn weevil, without using chemical insecticides.

6. If he continues to use insecticides, there are a number of harmful side effects of these chemicals that could harm the ecosystem. Describe **three** of these.